



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 2

**CANDIDATE
NAME**

--

CLASS

2T

**INDEX
NUMBER**

--	--	--	--

BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03
23RD AUGUST 2017
2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the Question paper.

Section B

Answer **one** question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
SECTION A	
1 [30]	
2 [12]	
3 [8]	
SECTION B	
4 [25]	
OR	
5 [25]	
TOTAL P3 [35%]	75

This document consists of **14** printed pages.

[Turn over

Section A

Answer **all** questions in this section.

- 1 Milk is an important source of diet for most infant. For many adults, milk is an important source of dietary calcium. Milk contains many biological molecules; one of them is lactose. Therefore, it is not surprising that most infant will have a mechanism to digest lactose. While for adults, it is not normal for their body to be able to digest lactose. However, there is an increasing trend of lactose tolerant (also known as lactase persistent) individual in the adult population. Here, we will discuss 3 types of conditions; namely Congenital Lactase Deficiency (CLD), lactose allergy, and lactose tolerance (lactase persistence).

Silanikove et al The Interrelationships between Lactose Intolerance and the Modern Dairy Industry: Global Perspectives in Evolutional and Historical Backgrounds Nutrients 2015, 7, 7312-7331

- (a) Fig. 1.1 shows the structure of lactose.

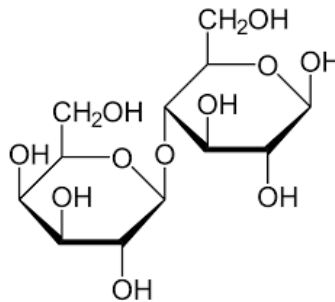


Fig. 1.1

Name the bond joining the 2 monomers in lactose

..... [1]

- (b) Fig. 1.2 shows the catalytic residues found in the active site of lactase.

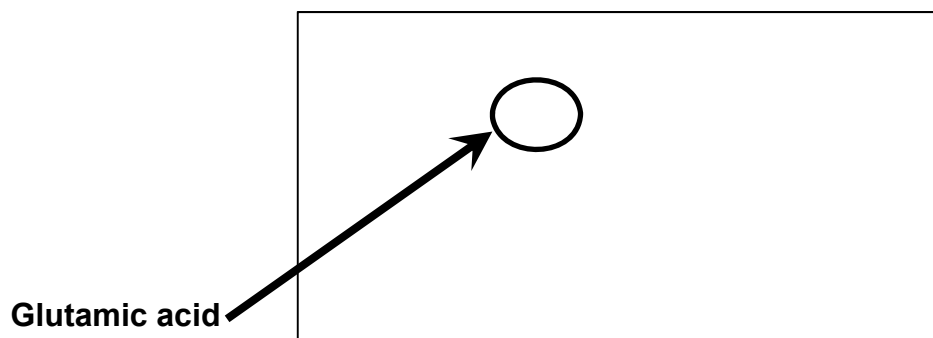


Fig. 1.2

One of the catalytic residues; glutamic acid (circled in Fig. 1.2) is substituted by glycine which is shown in Fig.1.3 below.

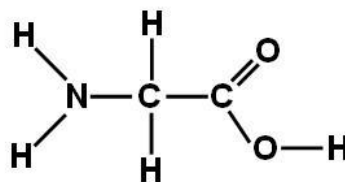


Fig. 1.3

Explain how lactase catalytic activity is affected by the substitution above.

.....

.....

.....

.....[2]

- (c)** Lactose intolerance in infant is also known as Congenital Lactase Deficiency (CLD). It is an autosomal recessive disorder.

Studies have shown that CLD is caused by mutation in *LCT* gene coding for lactase. The most commonly observed mutation is a single nucleotide substitution in *LCT* gene which results in the production of truncated lactase.

Other mutation such as a single nucleotide deletion in *LCT* gene has also been detected in a few patients which also results in the production of truncated lactase.

- (c)(i)** Explain how two different types of mutations; single nucleotide substitution and single nucleotide deletion in *LCT* gene can lead to the production of truncated lactase.

.....

.....

.....

..... [2]

- (c)(ii)** A small amount of DNA is isolated from infants suffering CLD resulted from a single nucleotide substitution. The DNA was subjected to process **Y** to ensure enough DNA for the subsequent Southern Blotting process.

Name the process **Y**.

..... [1]

Fig. 1.4 shows wild type *LCT* gene and mutant *LCT* gene. The mutation result in the loss of *NdeI* restriction site (*NdeI* RE) at position 4 kb as shown by the arrow ⁵↑.

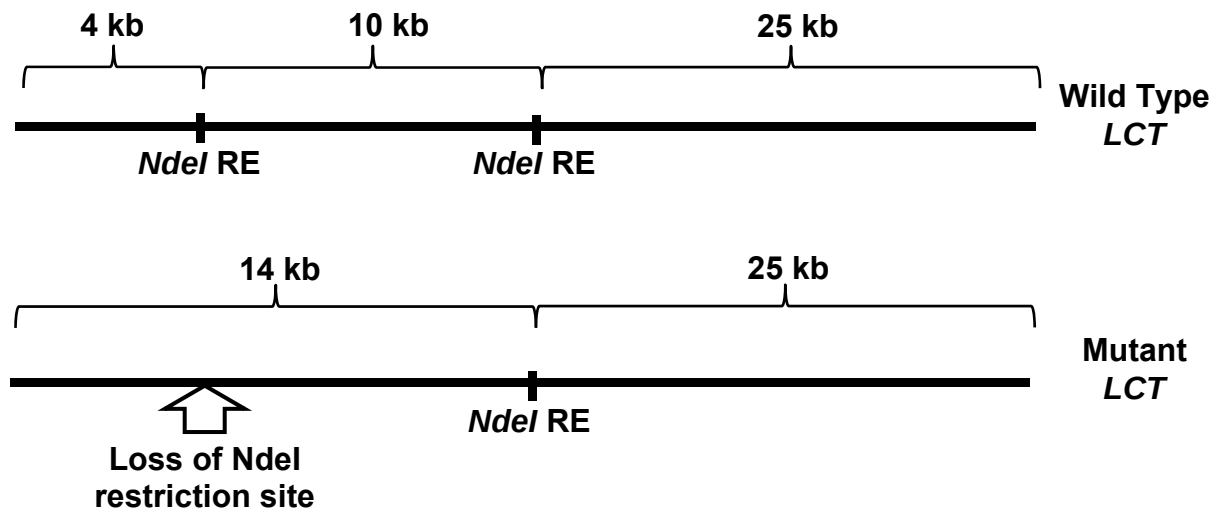


Fig. 1.4

Fig. 1.5 shows the band patterns of the nitrocellulose membrane obtained from the Southern Blotting of DNA sample from 3 infants.

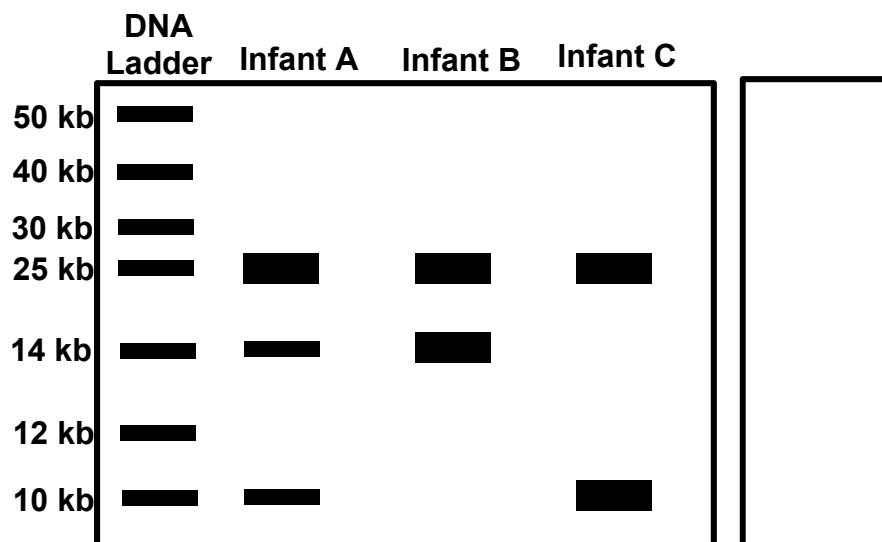
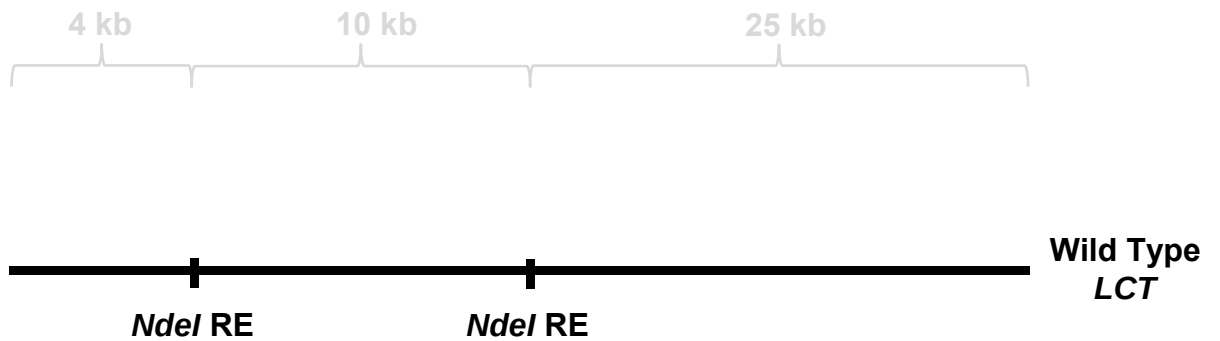


Fig. 1.5

(c)(iii) On the box on the right side of Fig. 1.5, indicate the position of the positive terminal and negative terminal during the gel electrophoresis which results in the band pattern in Fig. 1.5. [1]

- (c)(iv) Based on the position of the *NdeI* restriction sites in wild type and mutant *LCT* gene in Fig. 1.4 as well as the band patterns in Fig.1.5, indicate on the wild type *LCT* below where the probe will anneal to (use a ruled line and label). [1]



- (c)(v) Based on the information provided in part (c) as well as Fig. 1.4 and Fig. 1.5; explain which infant is suffering from CLD.

.....

.....

.....

..... [2]

- (d) Another condition known as lactose allergy results in more severe symptoms than lactose intolerance. The symptoms of allergy are due to the action of Immunoglobulin E (IgE) which activates mast cells, which subsequently secrete a chemical signal **X**.

- (d)(i) Name **X**.

..... [1]

Fig. 1.6 shows the structure of IgG and IgE.

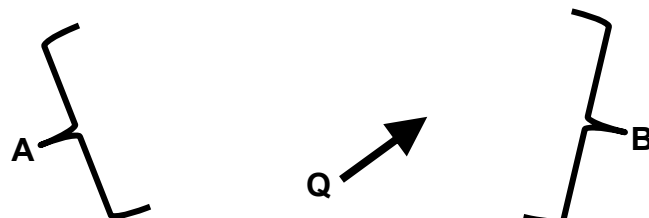


Fig. 1.6

- (d)(ii) The number, type, and position of Q on IgG and IgE are different. Name precisely the process which attaches Q on IgG and IgE.

- (d)(iii) Describe structures of IgE that allow it to perform its role in eliciting allergy response towards lactose.

.....

.....

.....

..... [2]

- (d)(iv) With reference to Fig. 1.6; suggest what will happen to an individual with lactose allergy when part **B** of all his IgE is replaced with part **A** of his IgG

.....

..... [1]

Some human adults continue to produce the lactase enzyme throughout their adulthood (lactase persistent). Therefore, they are able to digest lactose effectively (lactose tolerant) and will not develop symptoms such as bloating, flatulence, or diarrhoea after consuming milk.

However, most adult mammals stop producing the lactase enzyme (lactase non-persistent). Therefore, they are unable to digest lactose effectively (lactose intolerant) and will usually develop symptoms such as bloating, flatulence, or diarrhoea when consuming milk.

Fig. 1.7 shows the pattern of inheritance of lactose intolerance in Family A.

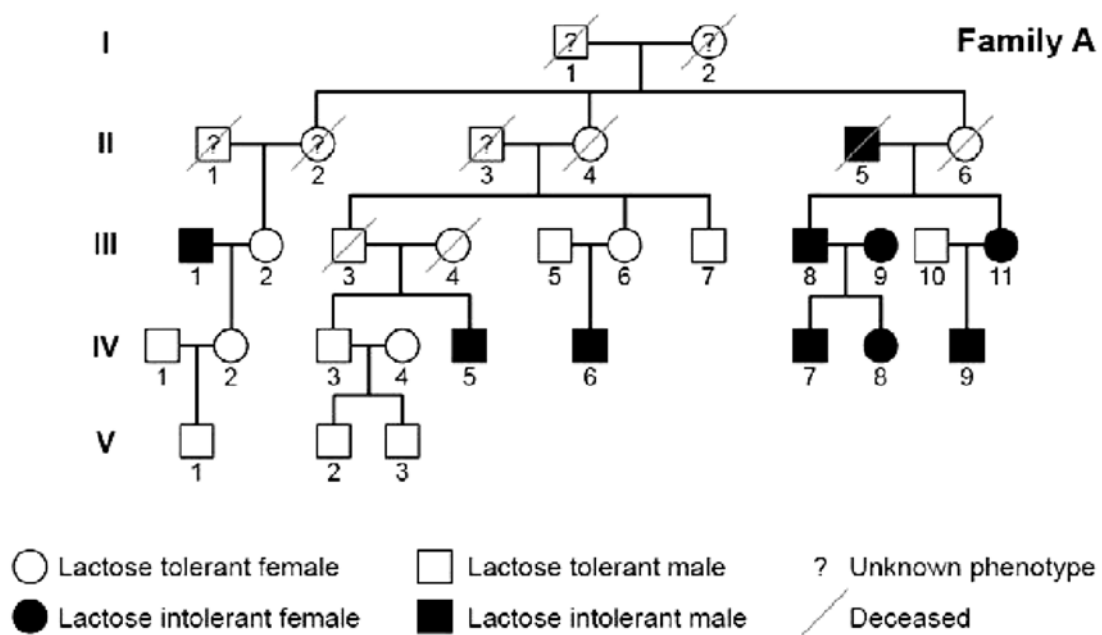


Fig. 1.7

- (e)(i) With reference to Fig. 1.7; explain the mode of inheritance of lactose intolerance and where the gene is probably located. Provide evidence to support your claim.

.....

.....

.....

.....

.....

.....

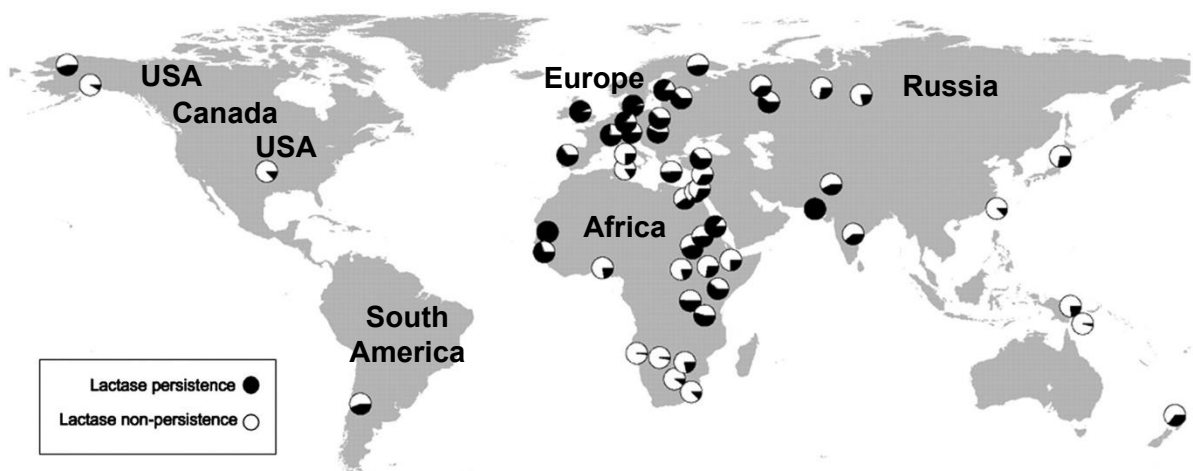
.....

.....[4]

Humans learned to exploit ruminants as a source of milk about 10,000 years ago; particularly the European population. Since then, the use of domesticated ruminants as a source of milk and dairy products has expanded until today when the dairy industry has become one of the largest sectors in the modern food industry, including the spread at the present time to countries such as China and Japan.

Widespread lactose intolerance among the adult population is a considerable drawback to dairy-based foods consumption. Over the centuries, three factors allowed humans to overcome limitations imposed by lactose intolerance: (i) mutations, which occurred in particular populations, most notably in the north European Celtic societies and African nomads, in which carriers of the lactose intolerance gene converted from being lactose intolerant to lactose tolerant; (ii) the ability to develop low-lactose products such as cheese and yogurt; and (iii) colon microbiome adaptation, which allow lactose intolerant individuals to overcome its intolerance.

Fig. 1.8 shows the pockets of lactase persistence shown in pie charts.



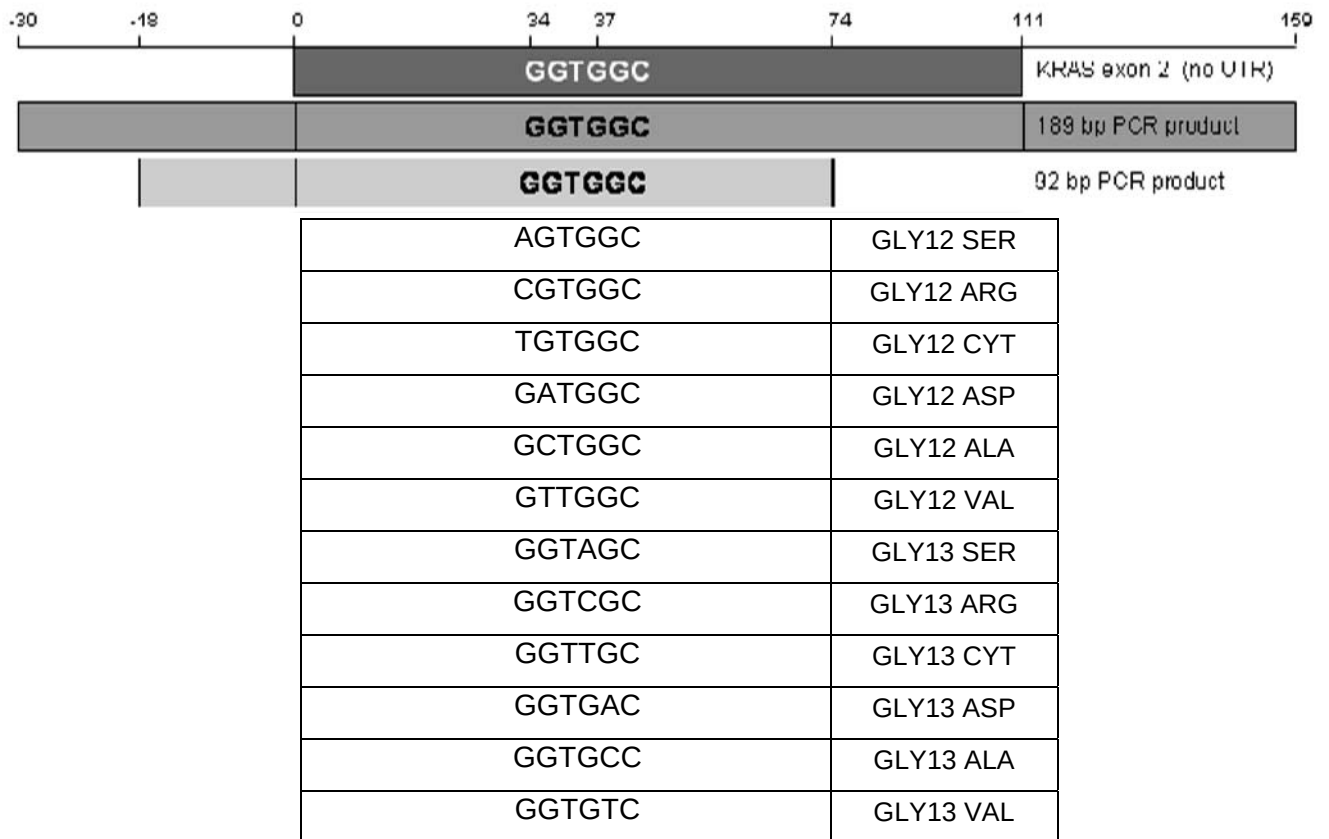
Pockets of lactase persistence shown in pie charts. Ingram et al, Hum Genet 124:579–591, 2009.

Fig. 1.8

- (e)(ii)** With reference to Fig. 1.8; apart from the genetic factors, suggest what other factor could have contributed to the spread of lactase persistence.
-
-
-
-[2]
- (f)(i)** Explain how Darwin's principles of evolution may be applied in understanding the type of evolution that would have had to take place in the spread of lactose tolerance.
-
-
-
-
-
-[3]
- (f)(ii)** Suggest how the scenario described in **(f)(i)** is an example of macro or micro evolution.
-
-
-
-[2]
- (g)(i)** Define anthropomorphic climate change.
-
-[1]
- (g)(ii)** Explain how the above scenario of increased lactase persistence may contribute to anthropomorphic climate change.
-
-
-
-

.....[3]

- 2 All mammalian cells express three closely related Ras proteins: H-Ras, K-Ras and N-Ras that promote oncogenesis when mutationally activated at codons 12, 13 or 61. Despite a high degree of similarity between the isoforms, K-Ras mutations are far more frequently observed in cancer. [Total: 30]



Location of K-RAS codon 12 and 13 mutations and PCR amplicons / products. Exon 2 of K-RAS is shown from the ATG without the untranslated region. The position and size of the PCR amplicons used in the High Resolution Melting HRM assays in relation to exon 2 of K-RAS is indicated. All possible mutations at codon 12 and 13 are listed along with the corresponding amino acid changes from Glycine (GLY) are shown.

Krypup et al. High resolution melting analysis for the rapid and sensitive detection of mutations in clinical samples: KRAS codon 12 and 13 mutations in non-small cell lung cancer BMC Cancer 2006, 6:295

Fig. 2.1

- (a) With reference to Fig. 2.1; explain the type of mutation experienced in K-RAS codon 13.

.....[2]

- (b) With reference to Fig. 2.1; suggest why oncogenesis in codon 13 is caused by only 6 possible changes in amino acid and not more.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

Ras proteins are the products of proto-oncogenes that are frequently mutated in human cancers. They are encoded by three ubiquitously expressed genes: *H-Ras*, *K-Ras* and *N-Ras*. These proteins are GTPases that function as molecular switches regulating pathways responsible for proliferation and cell survival.

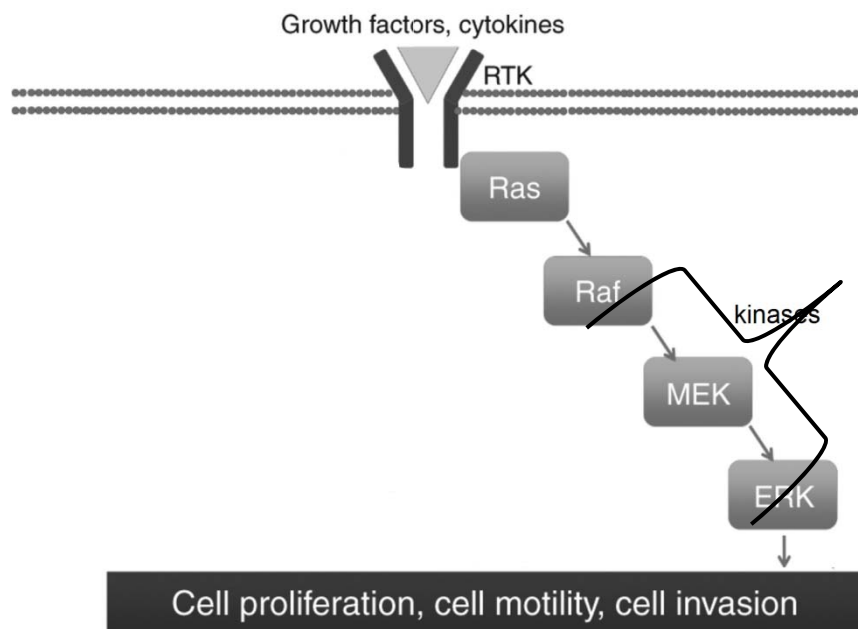


Fig. 2.2

- (c) How does Ras protein act as a molecular switch in initiating cell proliferation?

.....
.....[1]

- (d)** With reference to Fig. 2.2; explain how Ras protein is normally used to terminate cell proliferation.

.....
.....
.....
.....[2]

- (e)** With reference to Fig. 2.2; suggest how a mutation in *Ras* gene results in uncontrolled cell division.

.....
.....
.....
.....[1]

- (f)** With reference to **(a)** to **(e)**; explain the development of cancer.

.....
.....
.....
.....[2]

[Total: 12]

- 3** Cyanobacteria are a group of bacteria that obtains their energy through photosynthesis. They carry an operon known as *phycocyanin* operon which controls the expression of phycocyanin. Phycocyanin is a protein complex which serves as accessory pigment to chlorophyll in cyanobacteria. Without phycocyanin, light harvesting process is halted. The amount of phycocyanin increases from very low level to high level in the presence of light.

Fig. 3.1 below shows the structure of *phycocyanin* operon in the Cyanobacterium Anacystis nidulans.

P	O	CPCB1	CPA1	Intergenic region	P	O	CPCB2	CPA2
---	---	-------	------	-------------------	---	---	-------	------

Legend:

P	:	promoter
O	:	operator
CPCB1 and CPA1	:	structural genes coding for β – subunit of phycocyanin
CPCB2 and CPA2	:	structural genes coding for α – subunit of phycocyanin

Fig. 3.1

- (a)** Compare the *lac* operon to the *phycocyanin* operon.

.....

.....

.....

.....

.....

.....[3]

- (b)** Describe how the CPA2 gene could be transferred to another bacterium by a prophage.

.....

.....

.....

.....

.....

.....[3]

- (c) Cyanobacterium Anacystis nidulans has thylakoid which contain the same photosystems as the thylakoid of plant cells. Fig. 3.2 shows a hybrid *phycocyanin* operon.

trp P	O	CPCB1	CPCA1	Intergenic region	trp P	O	CPCB2	CPCA2
-------	---	-------	-------	-------------------	-------	---	-------	-------

Legend:
trp P

: trp promoter

Fig. 3.2

Explain the rate of production of oxygen in the Cyanobacterium Anacystis nidulans carrying the hybrid operon when light is present and tryptophan is present.

.....

.....

.....

.....[2]

[Total: 8]

Section B

Answer **one** question in this section

Write your answers on separate answer paper provided.

Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4** **(a)** Explain the fluid mosaic model and the roles of the constituent biomolecules in functions of membranes at the cell surface and of membranes within the cell. [13]
- (b)** Explain how genetic variation arises in a natural population and its significance in allowing the population to adapt and evolve. [12]

[Total: 25]

- 5** **(a)** With reference to named examples, describe the roles of proteins in bringing about cell signalling in living organisms. [13]
- (b)** Gene expression in eukaryotes is regulated at many different stages of the process. Explain how gene expression is regulated in eukaryotes and the significance of this at each stage. [12]

[Total: 25]